

*Bayesian and path model diagrams*

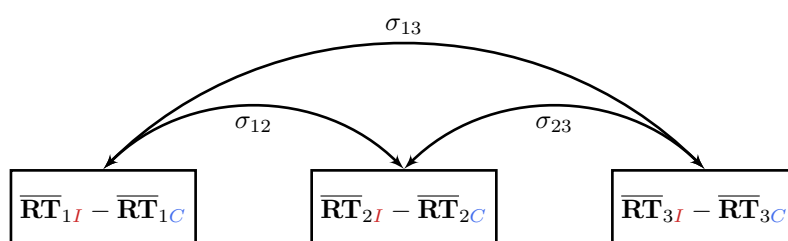
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**Generalized Hierarchical Factor Models**  
**Gaussian, ex-Gaussian and Shifted-lognormal**  
**Confirmatory and Exploratory model diagrams**

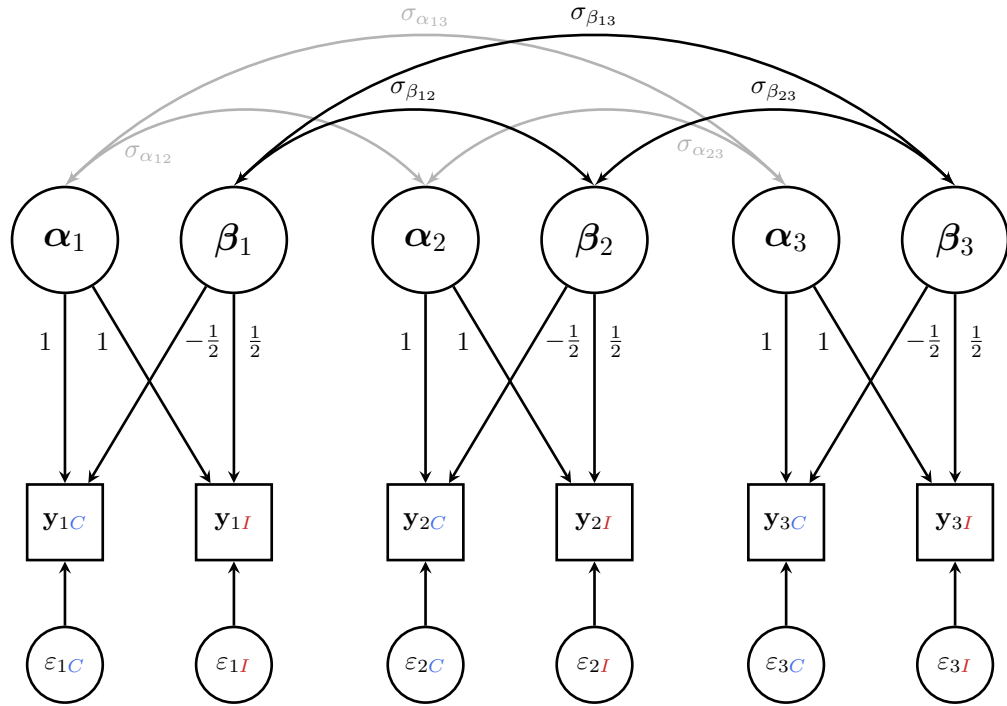
Ricardo Rey-Sáez, Alicia Franco-Martínez,  
Javier Revuelta, Miguel A. Vadillo.

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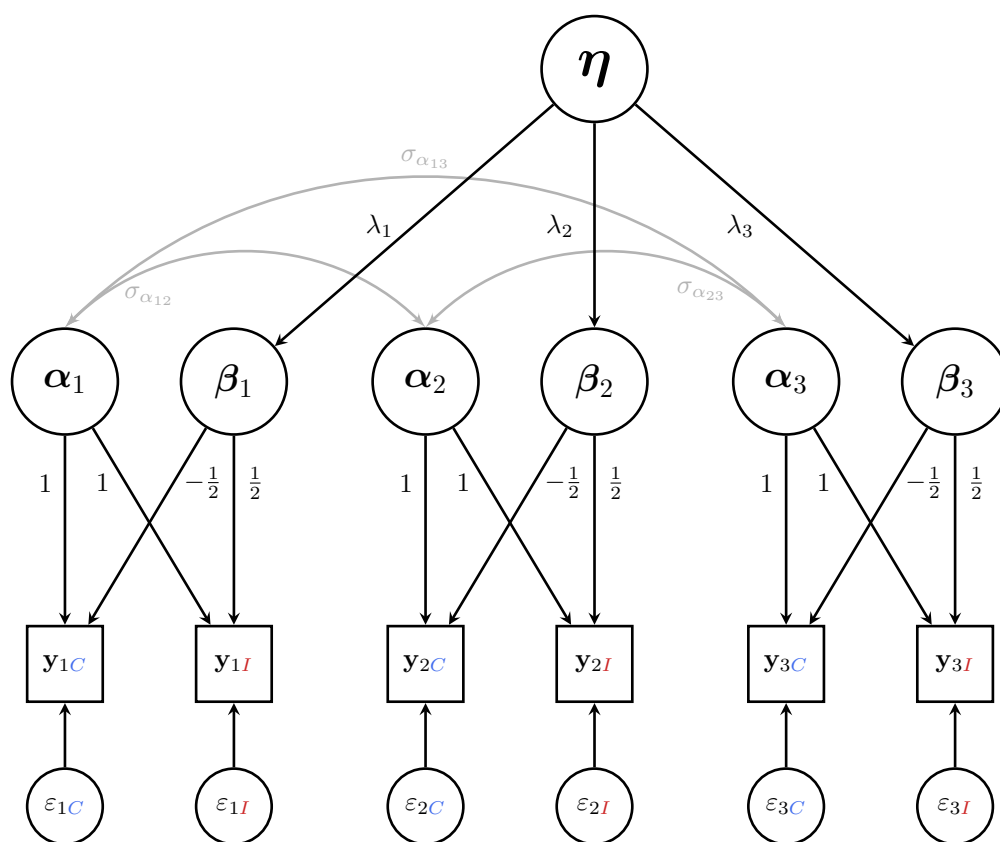
## Correlation between aggregated effects



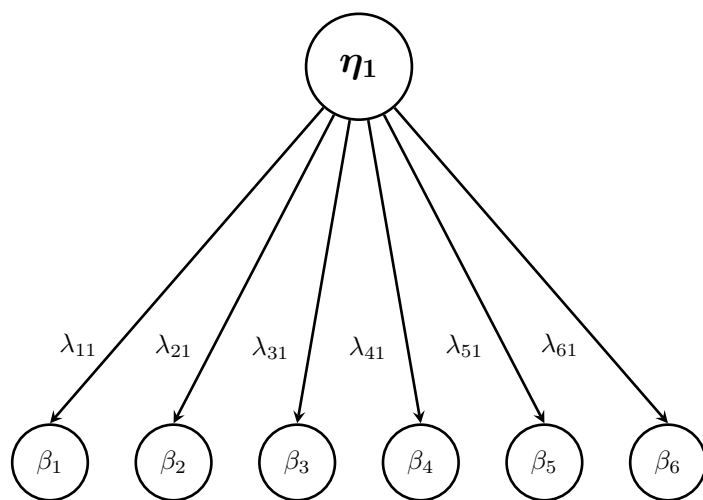
# Hierarchical models



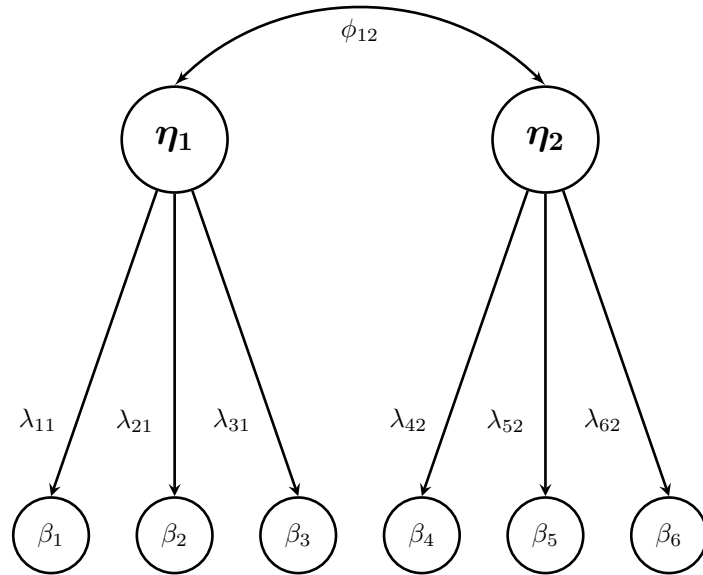
## Hierarchical factor models



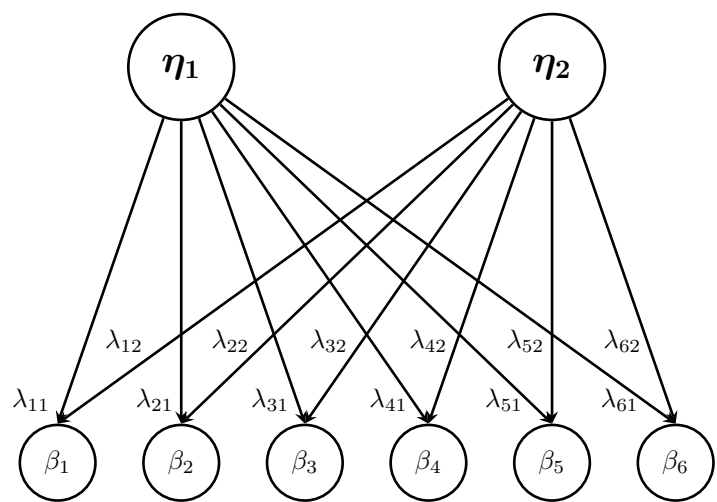
## Unidimensional model



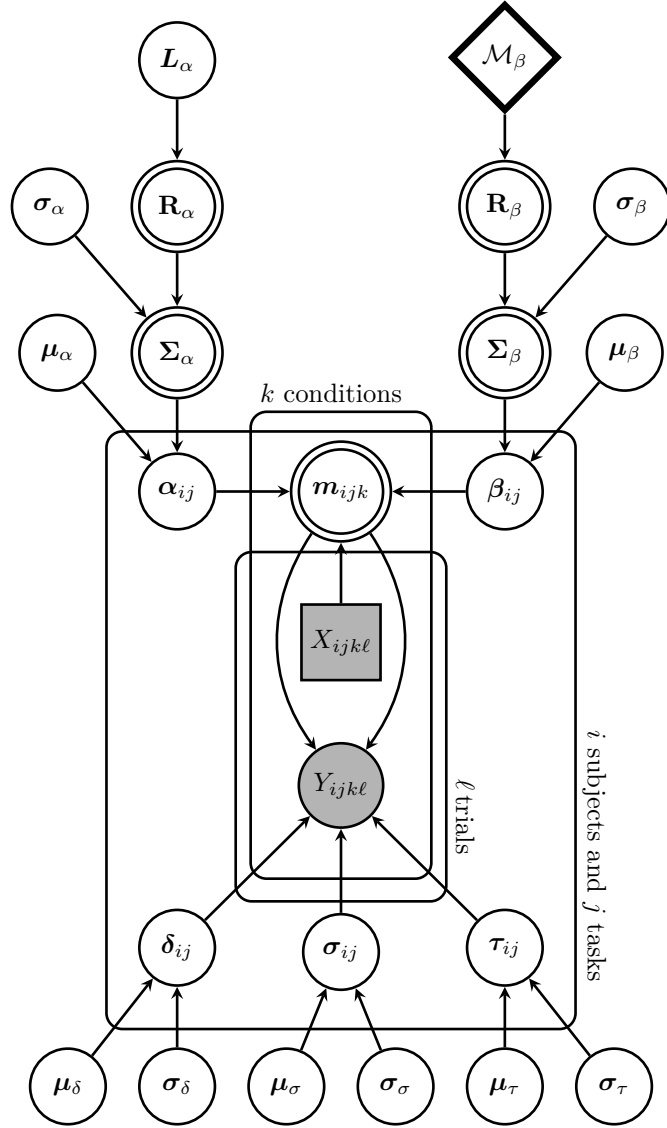
## Confirmatory factor model



Exploratory factor model



# Generalized HFM

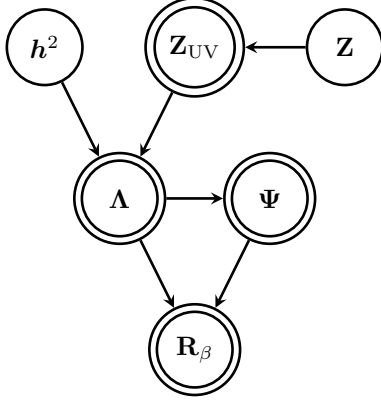




## GenHFM priors

$$\begin{array}{lll}
\boldsymbol{\mu}_\alpha & \sim & \text{Normal}(\mu_{\mu_\alpha}, \sigma_{\mu_\alpha}) \\
\boldsymbol{\mu}_\beta & \sim & \text{Normal}(\mu_{\mu_\beta}, \sigma_{\mu_\beta}) \\
\boldsymbol{\mu}_\sigma & \sim & \text{Normal}^+(\mu_{\mu_\sigma}, \sigma_{\mu_\sigma}) \\
\boldsymbol{\mu}_\tau & \sim & \text{Normal}^+(\mu_{\mu_\tau}, \sigma_{\mu_\tau}) \\
\boldsymbol{\mu}_\delta & \sim & \text{Normal}^+(\mu_{\mu_\delta}, \sigma_{\mu_\delta})_{\text{T}[0, u_j^{(1)}]} \\
\boldsymbol{\sigma}_\alpha & \sim & \text{Student-t}^+(\nu_{\sigma_\alpha}, \mu_{\sigma_\alpha}, \sigma_{\sigma_\alpha}) \\
\boldsymbol{\sigma}_\beta & \sim & \text{Student-t}^+(\nu_{\sigma_\beta}, \mu_{\sigma_\beta}, \sigma_{\sigma_\beta}) \\
\boldsymbol{\sigma}_\sigma & \sim & \text{Student-t}^+(\nu_{\sigma_\sigma}, \mu_{\sigma_\sigma}, \sigma_{\sigma_\sigma}) \\
\boldsymbol{\sigma}_\tau & \sim & \text{Student-t}^+(\nu_{\sigma_\tau}, \mu_{\sigma_\tau}, \sigma_{\sigma_\tau}) \\
\boldsymbol{\sigma}_\delta & \sim & \text{Student-t}^+(\nu_{\sigma_\delta}, \mu_{\sigma_\delta}, \sigma_{\sigma_\delta}) \\
\boldsymbol{L}_\alpha & \sim & \text{LKJ}(\eta_{L_\alpha}) \\
\mathbf{R}_\alpha & = & \boldsymbol{L}_\alpha \cdot \boldsymbol{L}'_\alpha \\
\mathbf{R}_\beta & = & \mathcal{M}_\beta \\
\boldsymbol{\Sigma}_\alpha & = & \text{diag}(\boldsymbol{\sigma}_\alpha) \cdot \mathbf{R}_\alpha \cdot \text{diag}(\boldsymbol{\sigma}_\alpha) \\
\boldsymbol{\Sigma}_\beta & = & \text{diag}(\boldsymbol{\sigma}_\beta) \cdot \mathbf{R}_\beta \cdot \text{diag}(\boldsymbol{\sigma}_\beta) \\
\boldsymbol{\alpha}_{ij} & \sim & \text{MVNormal}(\boldsymbol{\mu}_\alpha, \boldsymbol{\Sigma}_\alpha) \\
\boldsymbol{\beta}_{ij} & \sim & \text{MVNormal}(\boldsymbol{\mu}_\beta, \boldsymbol{\Sigma}_\beta) \\
\boldsymbol{\sigma}_{ij} & \sim & \text{Normal}^+(\boldsymbol{\mu}_{\sigma_j}, \boldsymbol{\sigma}_{\sigma_j}) \\
\boldsymbol{\tau}_{ij} & \sim & \text{Normal}^+(\boldsymbol{\mu}_{\tau_j}, \boldsymbol{\sigma}_{\tau_j}) \\
\boldsymbol{\delta}_{ij} & \sim & \text{Normal}^+(\boldsymbol{\mu}_{\delta_j}, \boldsymbol{\sigma}_{\delta_j})_{\text{T}[0, u_{ij}^{(2)}]}
\end{array}
\quad
\begin{array}{l}
\boldsymbol{m}_{ijk} = \begin{cases} \boldsymbol{\alpha}_{ij} - \boldsymbol{\beta}_{ij} \cdot \frac{1}{2} & \text{if } X_{ijk\ell} = 1, \\ \boldsymbol{\alpha}_{ij} + \boldsymbol{\beta}_{ij} \cdot \frac{1}{2} & \text{if } X_{ijk\ell} = 2. \end{cases} \\
Y_{ijk\ell} \sim \begin{cases} \text{Normal}(\boldsymbol{m}_{ijk}, \boldsymbol{\sigma}_{ij}), \\ \text{ex-Gaussian}(\boldsymbol{m}_{ijk}, \boldsymbol{\sigma}_{ij}, \boldsymbol{\tau}_{ij}), \\ \text{Shifted-Lognormal}(\boldsymbol{m}_{ijk}, \boldsymbol{\sigma}_{ij}, \boldsymbol{\delta}_{ij}). \end{cases}
\end{array}$$

## GenHFM: Exploratory Factor model



$$\mathbf{h}^2 \sim \text{Beta}(\alpha_{\mathbf{h}^2}, \beta_{\mathbf{h}^2})$$

$$\mathbf{Z} \sim \text{RPG}(\xi)$$

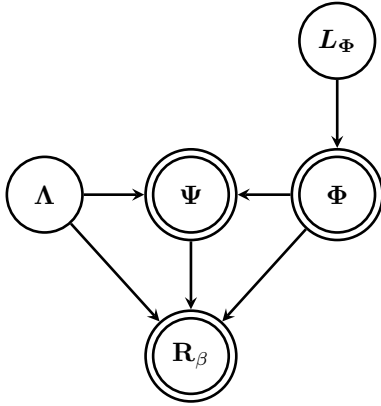
$$\mathbf{Z}_{\text{UV}} = \text{diag} \left[ (\mathbf{Z} \cdot \mathbf{Z}')_{jj} \right]^{-\frac{1}{2}} \cdot \mathbf{Z}$$

$$\mathbf{\Lambda} = \text{diag}(\mathbf{h}^2)^{\frac{1}{2}} \cdot \mathbf{Z}_{\text{UV}}$$

$$\mathbf{\Psi} = \mathbf{I} - \text{diag}(\mathbf{h}^2)$$

$$\mathbf{R}_{\beta} = \mathbf{\Lambda} \cdot \mathbf{\Lambda}' + \mathbf{\Psi}$$

## GenHFM: Confirmatory Factor model



$$\mathbf{\Lambda} \sim \text{Beta}_{[-1,1]}(\alpha_{\Lambda}, \beta_{\Lambda})$$

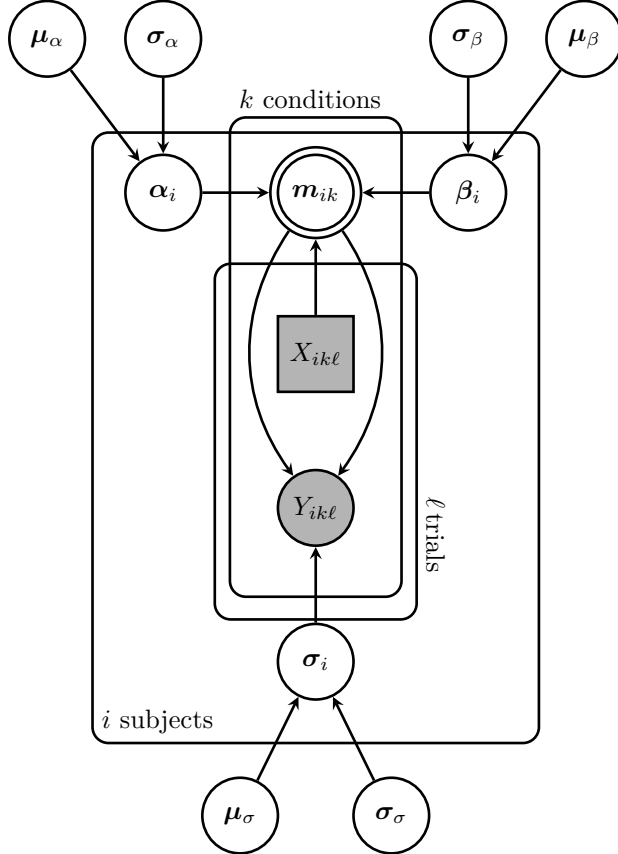
$$\mathbf{L}_{\Phi} \sim \text{LKJ}(\eta_{\mathbf{L}_{\Phi}})$$

$$\mathbf{\Phi} = \mathbf{L}_{\Phi} \cdot \mathbf{L}'_{\Phi}$$

$$\mathbf{\Psi} = \mathbf{I} - \text{diag}\left[(\mathbf{\Lambda} \cdot \mathbf{\Phi} \cdot \mathbf{\Lambda}')_{jj}\right]$$

$$\mathbf{R}_{\beta} = \mathbf{\Lambda} \cdot \mathbf{\Phi} \cdot \mathbf{\Lambda}' + \mathbf{\Psi}$$

## Gaussian Hierarchical Model



$$\mu_\alpha \sim \text{Normal}^+(\mu = 0, \sigma = 10)$$

$$\mu_\beta \sim \text{Normal}(\mu = 0, \sigma = 10)$$

$$\mu_\sigma \sim \text{Normal}^+(\mu = 0, \sigma = 10)$$

$$\sigma_\alpha \sim \text{Inverse-Gamma}(\alpha = 0.5, \beta = 0.5)$$

$$\sigma_\beta \sim \text{Inverse-Gamma}(\alpha = 0.5, \beta = 0.5)$$

$$\sigma_\sigma \sim \text{Inverse-Gamma}(\alpha = 0.5, \beta = 0.5)$$

$$\alpha_i \sim \text{Normal}(\mu = \mu_\alpha, \sigma = \sigma_\alpha)$$

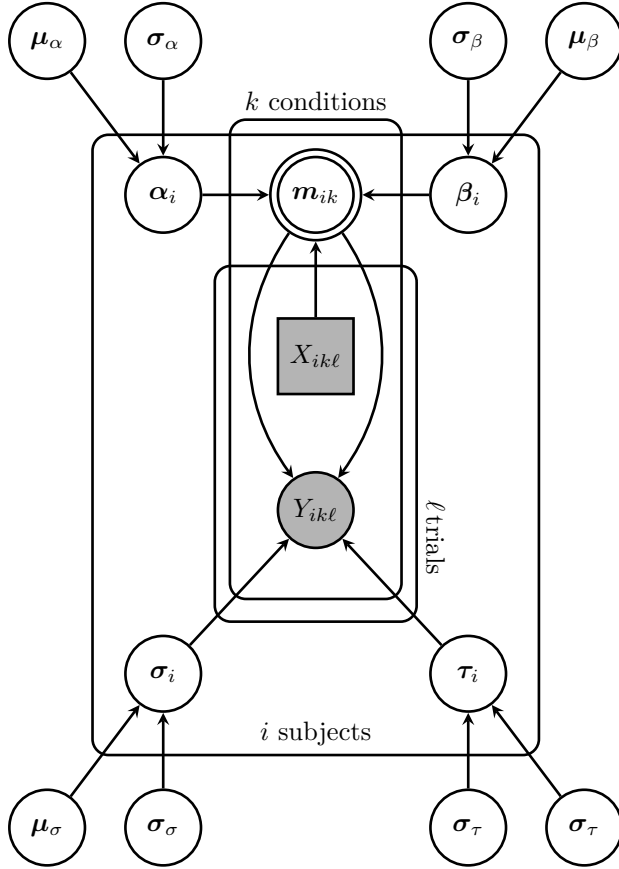
$$\beta_i \sim \text{Normal}(\mu = \mu_\beta, \sigma = \sigma_\beta)$$

$$\sigma_i \sim \text{Normal}^+(\mu = \mu_\sigma, \sigma = \sigma_\sigma)$$

$$\mathbf{m}_{ik} = \begin{cases} \alpha_i - \frac{1}{2} \cdot \beta_i & \text{if } X_{ik\ell} = 1 \\ \alpha_i + \frac{1}{2} \cdot \beta_i & \text{if } X_{ik\ell} = 2 \end{cases}$$

$$Y_{ik\ell} \sim \text{Normal}(\mu = \mathbf{m}_{ik}, \sigma = \sigma_i)$$

## Ex-Gaussian Hierarchical Model ( $\tau$ )



$$\mu_\alpha \sim \text{Normal}^+(\mu = 0, \sigma = 10)$$

$$\mu_\beta \sim \text{Normal}(\mu = 0, \sigma = 10)$$

$$\mu_\sigma \sim \text{Normal}^+(\mu = 0, \sigma = 10)$$

$$\mu_\tau \sim \text{Normal}^+(\mu = 0, \sigma = 10)$$

$$\sigma_\alpha \sim \text{Inverse-Gamma}(\alpha = 0.5, \beta = 0.5)$$

$$\sigma_\beta \sim \text{Inverse-Gamma}(\alpha = 0.5, \beta = 0.5)$$

$$\sigma_\sigma \sim \text{Inverse-Gamma}(\alpha = 0.5, \beta = 0.5)$$

$$\sigma_\tau \sim \text{Inverse-Gamma}(\alpha = 0.5, \beta = 0.5)$$

$$\alpha_i \sim \text{Normal}(\mu = \mu_\alpha, \sigma = \sigma_\alpha)$$

$$\beta_i \sim \text{Normal}(\mu = \mu_\beta, \sigma = \sigma_\beta)$$

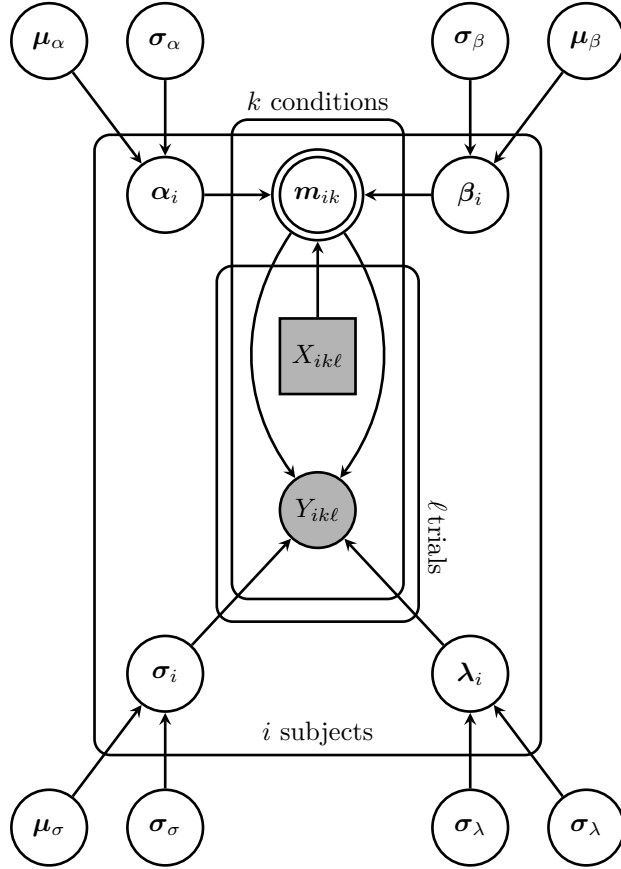
$$\sigma_i \sim \text{Normal}^+(\mu = \mu_\sigma, \sigma = \sigma_\sigma)$$

$$\tau_i \sim \text{Normal}^+(\mu = \mu_\tau, \sigma = \sigma_\tau)$$

$$\mathbf{m}_{ik} = \begin{cases} \alpha_i - \frac{1}{2} \cdot \beta_i & \text{if } X_{ikl} = 1 \\ \alpha_i + \frac{1}{2} \cdot \beta_i & \text{if } X_{ikl} = 2 \end{cases}$$

$$Y_{ikl} \sim \text{ex-Gaussian}\left(\mu = \mathbf{m}_{ik}, \sigma = \sigma_i, \lambda = \frac{1}{\tau_i}\right)$$

## Ex-Gaussian Hierarchical Model ( $\lambda$ )



$$\mu_\alpha \sim \text{Normal}^+(\mu = 0, \sigma = 10)$$

$$\mu_\beta \sim \text{Normal}(\mu = 0, \sigma = 10)$$

$$\mu_\sigma \sim \text{Normal}^+(\mu = 0, \sigma = 10)$$

$$\mu_\lambda \sim \text{Normal}^+(\mu = 0, \sigma = 10)$$

$$\sigma_\alpha \sim \text{Inverse-Gamma}(\alpha = 0.5, \beta = 0.5)$$

$$\sigma_\beta \sim \text{Inverse-Gamma}(\alpha = 0.5, \beta = 0.5)$$

$$\sigma_\sigma \sim \text{Inverse-Gamma}(\alpha = 0.5, \beta = 0.5)$$

$$\sigma_\lambda \sim \text{Inverse-Gamma}(\alpha = 0.5, \beta = 0.5)$$

$$\alpha_i \sim \text{Normal}(\mu = \mu_\alpha, \sigma = \sigma_\alpha)$$

$$\beta_i \sim \text{Normal}(\mu = \mu_\beta, \sigma = \sigma_\beta)$$

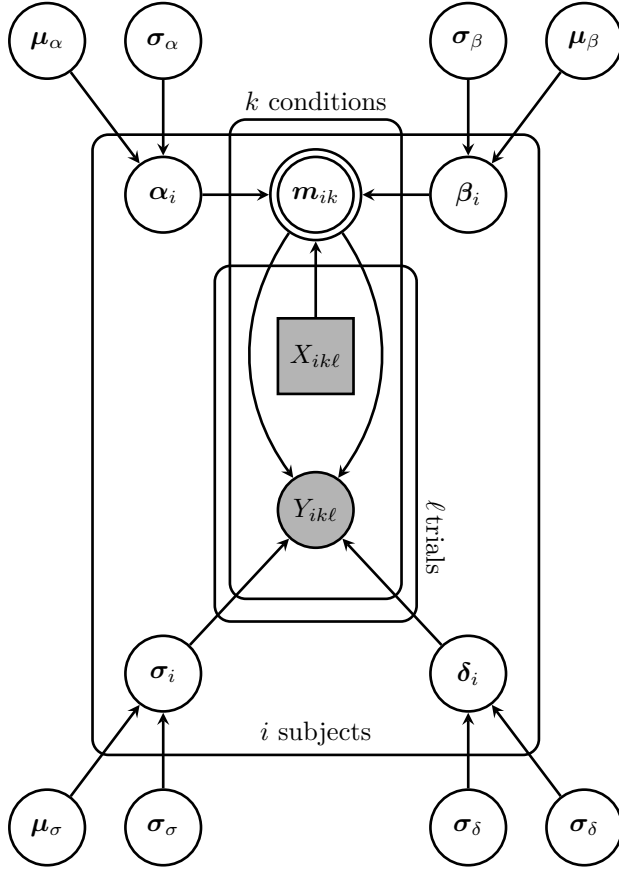
$$\sigma_i \sim \text{Normal}^+(\mu = \mu_\sigma, \sigma = \sigma_\sigma)$$

$$\lambda_i \sim \text{Normal}^+(\mu = \mu_\lambda, \sigma = \sigma_\lambda)$$

$$\mathbf{m}_{ik} = \begin{cases} \alpha_i - \frac{1}{2} \cdot \beta_i & \text{if } X_{ik\ell} = 1 \\ \alpha_i + \frac{1}{2} \cdot \beta_i & \text{if } X_{ik\ell} = 2 \end{cases}$$

$$Y_{ik\ell} \sim \text{ex-Gaussian}(\mu = \mathbf{m}_{ik}, \sigma = \sigma_i, \lambda = \lambda_i)$$

## Shifted-lognormal model



$$\mu_\alpha \sim \text{Normal}^+(\mu = 0, \sigma = 10)$$

$$\mu_\beta \sim \text{Normal}(\mu = 0, \sigma = 10)$$

$$\mu_\sigma \sim \text{Normal}^+(\mu = 0, \sigma = 10)$$

$$\mu_\delta \sim \text{Normal}^+(\mu = 0, \sigma = 10)$$

$$\sigma_\alpha \sim \text{Inverse-Gamma}(\alpha = 0.5, \beta = 0.5)$$

$$\sigma_\beta \sim \text{Inverse-Gamma}(\alpha = 0.5, \beta = 0.5)$$

$$\sigma_\sigma \sim \text{Inverse-Gamma}(\alpha = 0.5, \beta = 0.5)$$

$$\sigma_\delta \sim \text{Inverse-Gamma}(\alpha = 0.5, \beta = 0.5)$$

$$\alpha_i \sim \text{Normal}(\mu = \mu_\alpha, \sigma = \sigma_\alpha)$$

$$\beta_i \sim \text{Normal}(\mu = \mu_\beta, \sigma = \sigma_\beta)$$

$$\sigma_i \sim \text{Normal}^+(\mu = \mu_\sigma, \sigma = \sigma_\sigma)$$

$$\delta_i \sim \text{Normal}^+(\mu = \mu_\delta, \sigma = \sigma_\delta)$$

$$\mathbf{m}_{ik} = \begin{cases} \alpha_i - \frac{1}{2} \cdot \beta_i & \text{if } X_{ik\ell} = 1 \\ \alpha_i + \frac{1}{2} \cdot \beta_i & \text{if } X_{ik\ell} = 2 \end{cases}$$

$$Y_{ik\ell} \sim \text{Shifted-lognormal}(\mu = \mathbf{m}_{ik}, \sigma = \sigma_i, \delta = \delta_i)$$